Data Structures and Algorithms -3 (25CS2103E)

E-Commerce Product Entity-Resolution & Matching Platform

Team Members

2520090008-Sathish kumar

2520090071-Vignesh kumar

Under the Esteemed Guidance of

Dr. Anitha Chekuri
Assistant Professor

Contents

Abstract

- E-commerce websites contain multiple listings of the same product with different names, descriptions, and specifications.
- Our project identifies whether different listings represent the same real-world product.
- Product data is collected from multiple marketplaces and cleaned and standardized.
- Entity-resolution techniques are used to match products using brand, model, category, specifications, and features.
- Matched products are grouped into a single **canonical product**.
- The system removes duplicate records and enables accurate cross-marketplace price comparison.
- It improves product search, comparison, and overall shopping experience.

Introduction

- Platforms such as **Amazon, Flipkart, Google Shopping, PriceRunner, and Idealo** already provide product search and price comparison features.

- However, product information can be different across marketplaces.

- For example:
- **Amazon:** Apple iPhone 15 128GB Black
  Flipkart: iPhone 15 Black 128 GB

- Although the names are different, both listings may represent the same product.

- Our project focuses on the underlying problem of **identifying and matching these product entities**.

- The system creates a unified product catalog from different marketplace listings.

Literature Survey

Existing Platform/System	Main Features	Limitation / Difference
Amazon	Product search, seller listings, recommendations	Mainly operates within its marketplace
Flipkart	Product search, seller comparison, recommendations	Mainly operates within its marketplace
Google Shopping	Finds products from multiple online stores	Focuses mainly on shopping search and comparison
PriceRunner	Price comparison across sellers	Mainly focused on price comparison
Idealo	Product and price comparison	Consumer-oriented comparison platform
DataWeave	Product matching, retail analytics, price monitoring	Enterprise-oriented solution
Our Project	Entity resolution, duplicate detection, matching, unified catalog, price comparison	DSA-focused implementation with explainable matching

Problem Statement and Objectives

Problem Statement

Different e-commerce marketplaces may list the same product using different names, formats, descriptions, and specifications. This creates duplicate product records and makes accurate product comparison difficult.

Objectives

1. Collect product information from multiple marketplaces.
2. Clean and standardize product data.
3. Identify duplicate and similar product listings.
4. Match identical products using multiple attributes.
5. Calculate a product matching/confidence score.
6. Create a unified canonical product catalog.
7. Compare prices from different sellers.
8. Identify specification differences between products.
9. Provide the best available deal to the customer.
10. Improve product search accuracy.

Results

- Successfully identify duplicate product listings.
- Match products with different names but similar specifications.
- Generate a confidence score for product matching.
- Group identical listings into one canonical product.
- Display different sellers for the same product.
- Compare prices between sellers.
- Identify differences in product specifications.
- Improve search accuracy by reducing duplicate results.

Hardware and Software requirements

Hardware

- Windows Laptop/PC
- Intel i7
- 4 GB RAM or above
- 10 GB free storage
- Internet connection

Software

- Windows 10/11
- HTML5, CSS3, JavaScript
- Java
- Visual Studio Code
- Git & GitHub
- DSA Algorithms

Conclusion

- The proposed platform solves the problem of duplicate and inconsistent product listings across e-commerce marketplaces.
- It uses entity resolution to determine whether different listings represent the same product.
- Product information is cleaned, standardized, matched, and grouped into a unified catalog.
- The system can provide confidence scores and explain why products are considered a match.
- Cross-marketplace comparison makes it easier to identify the best available product and price.
- The project demonstrates practical applications of **Data Structures and Algorithms in e-commerce**.

Future Scope

- AI/ML-based Product Matching for more accurate entity resolution.
- Real-time Price Tracking across marketplaces.
- Personalized Product Recommendations based on user preferences.
- Fake/Suspicious Listing Detection.
- Seller Reliability Score based on ratings and history.
- Multilingual Product Matching for different languages.
- Image-Based Product Matching using product images.
- Real-Time Inventory Availability.
- Price History Analysis to identify the best time to purchase.

Thank you